DATABASES

ACID Transactions Visual Guide

Atomicity, Consistency, Isolation, Durability -- and the machinery that delivers them.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Atomicity & Durability

All-or-nothing execution, then surviving a crash.

```
-- durability: COMMIT appends to the Write-Ahead Log first
-- crash recovery replays the WAL: committed txns restored,
-- uncommitted txns rolled back. No torn writes.
```

Write-Ahead Log (WAL)

append-only log of every change -- the source of truth after a crash

STEP 2/6

Read Phenomena

The three anomalies isolation levels defend against.

Dirty Read

read UNCOMMITTED data -- may be rolled back

Non-Repeatable Read

same row read twice, values differ

Phantom Read

re-run query, NEW rows appear in the range

Stronger isolation eliminates more phenomena -- at the cost of concurrency and throughput.

STEP 3/6

Isolation Levels

Four levels, from fastest to strictest.

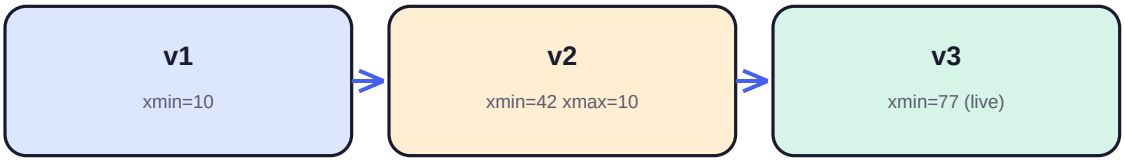
	Dirty read	Non-rep.	Phantom
READ UNCOMMITTED rarely used	yes	yes	yes
READ COMMITTED Postgres / Oracle default	no	yes	yes
REPEATABLE READ MySQL / InnoDB default	no	no	yes
SERIALIZABLE strictest, slowest	no	no	no

Green = phenomenon prevented. Red = phenomenon possible.

STEP 4/6

MVCC: Readers Never Block Writers

Every row carries versions; each transaction sees a snapshot.



UPDATE writes a NEW version; the old one stays until no snapshot needs it (VACUUM reclaims).

Snapshot Isolation

txn starting at T sees versions committed before T -- writers never block readers

Cost: bloat. Dead versions pile up under long-running transactions -- keep txns short.

STEP 5/6

Locking

When MVCC is not enough, locks serialize access.

Row locks

UPDATE / DELETE take them automatically; SELECT ... FOR UPDATE takes them explicitly

Table locks

DDL (ALTER TABLE) blocks writers -- the classic deploy outage

Advisory locks

application-level mutexes (pg_advisory_lock) -- no table involved

Deadlock

T1 holds A waits B;■T2 holds B waits A

→

Fix

lock in a fixed■order; keep txns short

STEP ✓/✓

Key Takeaways

ACID Transactions -- Remember

- Atomicity: COMMIT makes everything visible, ROLLBACK makes nothing visible.
- Durability comes from the WAL: committed data survives crashes via replay.
- Isolation is a spectrum: dirty reads -> non-repeatable reads -> phantoms, eliminated level by level.
- MVCC gives readers a snapshot so writers never block them -- at the cost of version bloat.
- Locks serialize what MVCC cannot; lock tables in a fixed order and keep transactions short.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev